Linwei Tao (Last updated: 4 September, 2026)

(+61) 0413002799
linwei.tao@sydney.edu.au
https://www.taolinwei.com

Education

School of Computer Science, University of Sydney Sydney, Australia
Ph.D., Computer Science 2023 - 2026

- Advisors: Chang Xu & Daochang Liu
- Research area: LLM Uncertainty Quantification and Confidence Calibration
- Expected thesis submission: 31 December 2026

School of Computer Science, University of Sydney Sydney, Australia
M.Phil., Computer Science 2021 - 2023

- Advisors: Chang Xu & Lijun Chang
- Research area: Uncertainty Quantification, Confidence Calibration, Neural Architecture Search

School of Computer Science, University of Sydney Sydney, Australia
M.E., Data Science 2020 - 2021

Huazhong University of Science and Technology Wuhan, China
B.E., Communication Engineering 2013 - 2017

- Received Science and Technology Scholarship

Experiences

Research Intern | Apple MLR, Cambridge UK 2026.05 - 2026.10

- Multi-turn RL post-training of LLM agents for interactive reasoning and tool use.
- Reward and environment design, tool calling, and rollout audits for reward hacking.
- Training efficiency: memory layout for agent communication, async vs. sync rollouts.

Research Intern | Google Research, Sydney AU 2025.11 - 2026.02

- Speculative decoding for efficient LLM inference.

Teaching Assistant | University of Sydney, Sydney 2022 - now

- COMP5329 Deep Learning, COMP5318 Machine Learning and Data Mining, OCMP5329 Deep Learning (Online), HTIN5005 Applied Healthcare Data Science, BUSS6002 Data Science in Business, QBUS5010 Dashboarding and Data Visualisation.

Construction Supervisor | Songjiang Construction Union, Shanghai 2017.07 - 2019.10
IOS Developer Intern | Trip.com, Shanghai 2015.06 - 2015.09
Frontend Developer Intern | Sunallies, Shanghai 2014.06 - 2014.09

Selected Publications

1. **Linwei Tao**, Yi-Fan Yeh, Minjing Dong, Tao Huang, Philip Torr, Chang Xu. Revisiting Uncertainty Estimation and Calibration of Large Language Models. *NeurIPS* 2025 Scaling Environments for Agents (SEA) Workshop.

2. **Linwei Tao**, Yi-Fan Yeh, Bo Kai, Minjing Dong, Tao Huang, Tom A. Lamb, Jialin Yu, Philip H.S. Torr, Chang Xu. Can Large Language Models Express Uncertainty Like Human? *Preprint*, 2026.

3. Younan Zhu, **Linwei Tao**, Minjing Dong, Chang Xu. Attention Calibration for Reducing Hallucination in Large Vision-Language Models. *Preprint*, 2026.

4. **Linwei Tao**, Younan Zhu, Haolan Guo, Minjing Dong, Chang Xu. A Benchmark Study on Calibration. *International Conference on Learning Representations, ICLR* 2024.

Awards and Grants

- **Apple Scholars in AI/ML PhD Fellowship** (20 selected worldwide, first Australian recipient), Apple 2026.04
- **FFT Student Survey Award for Outstanding Achievement in BUSS6002**, USYD Business School 2025.12
- **AAAI 2026 Student Program Committee** (3 selected worldwide), AAAI 2025.06
- **International Tuition Fee Scholarship**, USYD 2023.06
- **Faculty of Engineering Research Support Scholarship**, USYD 2023.05
- **Excellent Student Cadre**, HUST, 2015.09
- **Science and Technology Scholarship**, HUST 2014.06

Other Publications

1. Haolan Guo, **Linwei Tao**, Haoyang Luo, Minjing Dong, Chang Xu. Sample Margin-Aware Recalibration of Temperature Scaling. *International Conference on Machine Learning, ICML* 2026.
2. Xiaoyang Li, **Linwei Tao**, Haohui Lu, Minjing Dong, Junbin Gao, Chang Xu. Calibrating Graph Neural Networks with Wavelet-Aware Temperature Scaling. *International Conference on Learning Representations, ICLR* 2026.
3. Lei Pan, Zhipeng Lu, Yuan Zheng, Chongyao Yan, Haitao Wen, **Linwei Tao**, Chang Xu. Task-adaptive continual learning of vision language models via prototype routing and prompt. *Neurocomputing*, 2026.
4. Yingqing Yuan, **Linwei Tao**, Haohui Lu, Matloob Khushi, Imran Razzak, Mark Dras, Jian Yang, Usman Naseem. KG-UQ: Knowledge Graph-Based Uncertainty Quantification for Long Text in Large Language Models. *International World Wide Web Conference, WWW* 2025 Workshop on Sustainable AI for the Future Web
5. Haoyang Luo, **Linwei Tao**, Minjing Dong, Chang Xu. Beyond One-Hot Labels: Semantic Mixing for Model Calibration. *International Conference on Machine Learning, ICML* 2025.
6. Jinxu Lin, **Linwei Tao**, Minjing Dong, Chang Xu. Uncertainty Weighted Gradients for Model Calibration. *The IEEE / CVF Computer Vision and Pattern Recognition Conference, CVPR* 2025.
7. Jinxu Lin, **Linwei Tao**, Minjing Dong, Chang Xu. Diffusion Attribution Score: Evaluating Training Data Influence in Diffusion Model. *International Conference on Learning Representations, ICLR* 2025. (Spotlight, 5.1% of 11,500 submissions)
8. **Linwei Tao**, Minjing Dong, Chang Xu. Feature Clipping for Uncertainty Calibration. *AAAI Conference on Artificial Intelligence, AAAI* 2025.
9. Lei Pan, Wuyang Luan, Yuan Zheng, Junhui Li, **Linwei Tao**, Chang Xu. GraphFusion: Integrating Multi-Level Semantic Information with Graph Computing for Enhanced 3D Instance Segmentation. *Neurocomputing*, 2024.
10. **Linwei Tao**, Minjing Dong, Chang Xu. Dual Focal Loss for Calibration. *International Conference on Machine Learning, ICML* 2023.
11. **Linwei Tao**, Minjing Dong, Daochang Liu, Changming Sun, Chang Xu. Calibrating a Deep Neural Network with Its Predecessors. *International Joint Conference on Artificial Intelligence, IJCAI* 2023.

Preprints

1. Yunke Wang, **Linwei Tao**, Bo Du, Yutian Lin, Chang Xu. Visual Imitation Learning with Calibrated Contrastive Representation.
2. **Linwei Tao**, Haoyang Luo, Minjing Dong, Chang Xu. Confidence Calibration under Ambiguous Ground Truth. *arXiv:2603.22879*, 2026.
3. **Linwei Tao**, Haolan Guo, Minjing Dong, Chang Xu. Consistency Calibration: Improving Uncertainty Calibration via Consistency among Perturbed Neighbors.
4. Rui Xu, Yunke Wang, **Linwei Tao**, Wenjie Xuan, Yong Luo. Diversifying Personalized Research Ideation against AI-Induced Homogenization. *arXiv:2607.28087*, 2026.

Academic Services

Conference Reviewer: *ICML, NeuIPS, ICLR, CVPR, ICCV, AAAI, IJCAI, ACMMM, AISTATS*

Journal Reviewer: *T-MM, TMLR, DAMI, TPAMI, Machine Learning*